

Simple symmetric Venn diagrams with 17 and 19 curves

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September 21, 2026 (draft 4)

Abstract

We exhibit simple, rotationally symmetric Venn diagrams with 17 curves and with 19 curves: n Jordan curves carried to one another by rotation through $2\pi/n$, with every one of the 2^n regions present and connected and, since the diagrams are simple, every crossing on exactly two curves. Symmetric Venn diagrams exist for every prime number of curves (Griggs, Killian and Savage, 2004), but those diagrams have many curves through a point; simple ones were known only up to 13 curves (Mamakani and Ruskey, 2014). Four 17-curve and six 19-curve diagrams were found by a Metropolis walk on rotation-invariant quadrangulations of the sphere in which regions may temporarily be duplicated, started from the Griggs–Killian–Savage diagram with its multiple crossings resolved. Every diagram is given by a machine-checkable certificate; one certificate of each size has been verified by a formal proof in Lean 4. All of the diagrams are non-monotone, which is why the crossing-sequence searches that found the 11- and 13-curve diagrams could not have found them.

1 Statement

An n -Venn diagram is a family of n Jordan curves in the plane such that for every subset S of the curves the set of points inside exactly the curves of S is nonempty and connected. It is *simple* if no point lies on three curves, and *symmetric* if a rotation by $2\pi/n$ maps the family to itself. Henderson observed in 1963 that a symmetric n -Venn diagram can exist only for prime n [1]; his argument had a gap, and the first complete proof of his theorem is due to Wagon and Webb [2]. Grünbaum asked in 1975 whether they exist for every prime and gave simple symmetric diagrams with five curves [3]; simple symmetric diagrams with seven curves were found by Grünbaum and by Edwards in the 1990s [4, 5]. Hamburger constructed the first symmetric 11-Venn diagram in 2002 [7]; it is not simple, and the same holds for the diagrams of Griggs, Killian and Savage, who then proved that symmetric n -Venn diagrams exist for every prime n [9] (see also [10]). Simple symmetric diagrams with 11 and 13 curves were found by Mamakani and Ruskey in 2012 and 2014 [13, 15] by a search over monotone diagrams, and the question for larger primes, put forward as the central open problem by Ruskey, Savage and Wagon [12], remained open: the recent literature lists simple symmetric n -Venn diagrams as known only for $n \leq 13$ [16]. The survey of Ruskey and Weston [11] has the full history.

Theorem 1. *Simple symmetric Venn diagrams with 17 curves exist, and so do simple symmetric Venn diagrams with 19 curves.*

The proof is by exhibition. Four 17-curve diagrams and six 19-curve diagrams are given as certificates in the form of Section 2, each accepted by two independently written checkers, and one of each size has been formally verified in Lean 4 (Section 3).

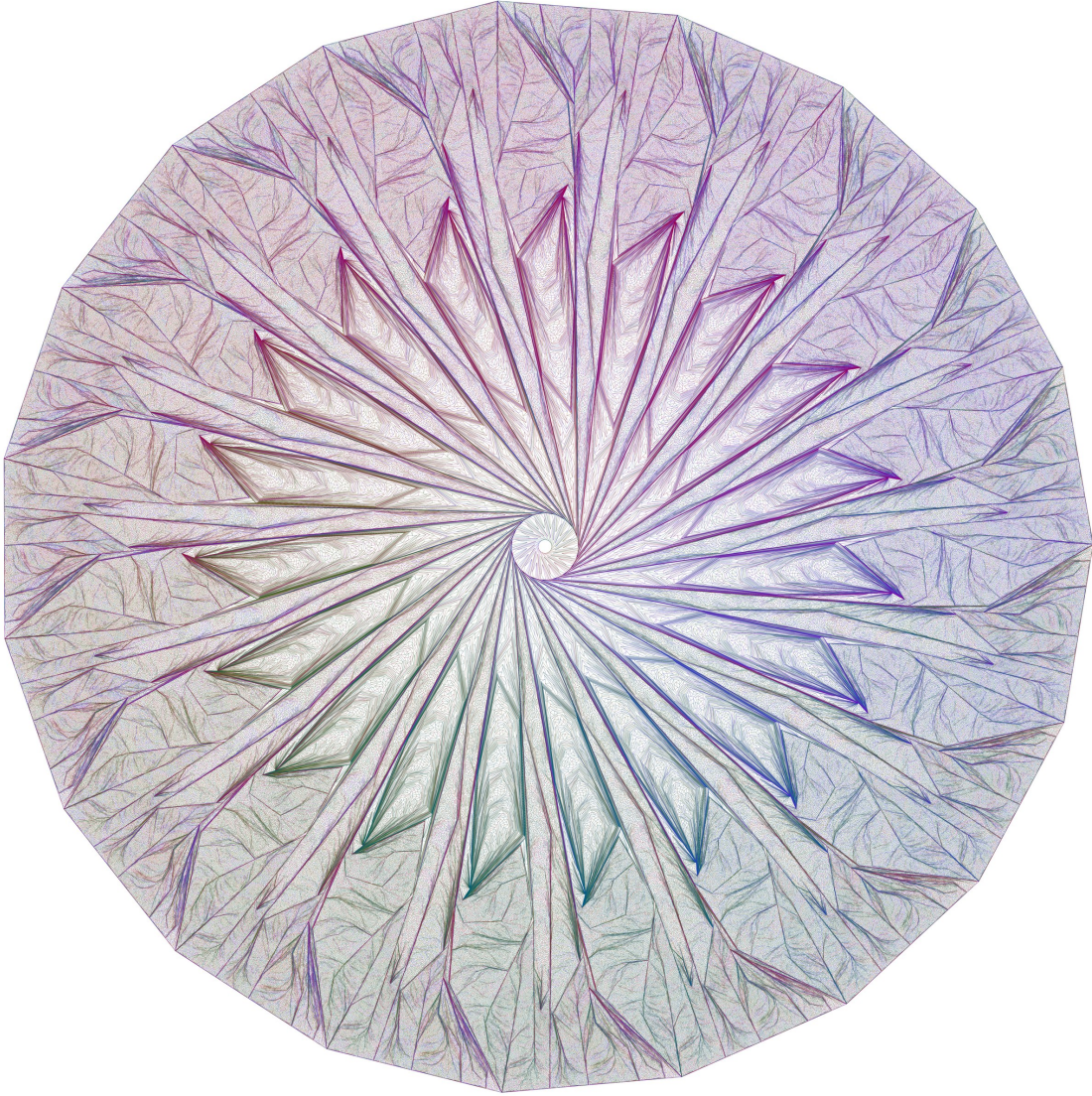


Figure 1: One of the six 19-curve diagrams (certificate `fc7ad34`), drawn from its certificate at near-uniform crossing density by the plotter described in the data repository: 19 curves, 524,286 crossings, every one of the 524,288 regions present exactly once. The centre hole is the region inside all 19 curves; the outer face is the region outside all of them. Colours identify curves.

2 Encoding and verification

Let D be a simple symmetric n -Venn diagram, drawn on the sphere with the unbounded region as a face. Its dual map M has one vertex per region, labelled by the region's membership pattern in $\{0, 1\}^n$; one edge per arc, joining labels that differ in exactly the bit of the arc's curve; and one face per crossing, a 4-cycle with labels $u, u \oplus e_a, u \oplus e_a \oplus e_b, u \oplus e_b$ (a *cube square*). Conversely:

Proposition 1. *A map on the sphere whose vertices are labelled by distinct elements of $\{0, 1\}^n$, all 2^n occurring, whose faces are cube squares, whose edges of each coordinate i form a single cycle through the faces containing them, and whose vertex rotations are single cycles, is the dual of a simple n -Venn diagram; if the label rotation $x \mapsto (x_1, \dots, x_{n-1}, x_0)$ is an orientation-preserving automorphism of the map, the diagram is symmetric.*

Proof sketch. The map is a cellular embedding on the sphere; its geometric dual is a 4-regular plane graph whose vertices are the faces (crossings). The edges of coordinate i , chained through opposite sides of each crossing, form one closed curve C_i ; regions of the arrangement are the vertices of M , and by the label condition each point of $\{0, 1\}^n$ is realized by exactly one region, which is a disk. Rotation of the labels acts as a rotation of the sphere fixing the two vertices 0^n and 1^n ; taking 0^n as the outer face gives a plane diagram with n -fold rotational symmetry. $\square$

A certificate is the list of the $2^n - 2$ faces of M , each as four n -bit labels: 131,070 faces for $n = 17$ and 524,286 for $n = 19$. Verification recomputes, in seconds: all 2^n labels present exactly once, every edge in exactly two faces ($2^{n+1} - 4$ edges), Euler characteristic 2, every face transversal, a single rotation cycle at every vertex, one cycle per curve with both sides connected, and invariance under label rotation. Two independently written checkers, a half-edge implementation in C++ and a face-set implementation in Python, accept all ten certificates, as does a third, written later and sharing no code with either, which audits the raw search state directly.

n	certificate	SHA-256 prefix	non-isomorphic
17	venn17-local-c3-s2	c178d7bd	yes
17	venn17-gcp-s12	87f981b8	yes
17	venn17-gcp-s14	eec8cd55	yes
17	venn17-gcp-s16	431857aa	yes
19	venn19-closure-s196002	fc7ad34	yes
19	venn19-closure-s196004	285de8ea	yes
19	venn19-closure-s195001	a09ceae0	yes
19	venn19-closure-s196001	56e7e1a1	yes
19	venn19-closure-s196007	312e8c88	yes
19	venn19-closure-s195002	726d1fb0	yes

Table 1: The ten certificates. Within each size the diagrams are pairwise non-isomorphic under cyclic relabelling of the curves, mirror image and exchange of the two poles (canonical forms compared over all $4n$ such maps); as face sets the six 19-curve certificates differ pairwise by between 25,460 and 827,716 faces of 524,286.

3 Formal verification

The first 17-curve certificate (c178d7bd) was formally verified in Lean 4 with Mathlib by Justin Grimes [17], working from the definitions alone and without access to our code. His theorem states that the 17 curves constructed from the certificate are Jordan curves in the Euclidean plane, carried to one another by a rigid rotation through $2\pi/17$, with every one of the 2^{17} regions path-connected, no point on three curves, and every crossing a transversal crossing of two curves. The finite parts of the proof use Lean’s compiled evaluation (`native_decide`); the development builds from scratch in under an hour.

The first 19-curve certificate (fcb7ad34; JSON form ed26b3ba) was verified by a port of the same development to $n = 19$, carried out by Codex (OpenAI) from Grimes’s sources: the dimension-dependent constants were changed, the finite meridian and sector certificates that the proof consumes were regenerated for 19, and the vendored topology library was left untouched. The port builds in twenty minutes and proves `exists_simple_rotational_venn_19`: there exist $C : \text{Fin } 19 \rightarrow \mathcal{P}(\mathbb{R}^2)$ and sides S with `SimpleRotationalVenn` C S , the same predicate as at 17. Its axiom list is that of the 17-curve proof under the namespace rename: `propext`, `Classical.choice`, `Quot.sound` and the twenty `native_decide` certificates. We re-ran `#print axioms` on the built project to confirm this.

4 Method

We search the space of rotation-invariant cube-square quadrangulations of the sphere in which several vertices may carry the same label. The energy of a state is $E = (\text{missing labels}) + \lambda (\text{duplicate regions})$; $E = 0$ with $\lambda > 0$ is exactly a certificate. Moves are the local Reidemeister-type rewirings of the dual (lens insertion and removal on a 3-cube, triangle flip, bigon insertion and removal), each applied simultaneously to all n rotation images; every move preserves the sphere, the cube-square faces and the one-cycle-per-curve property, so the walk never leaves the space of valid symmetric arrangements. Proposals are accepted by the Metropolis rule at temperature T , with half of the proposals targeted at a missing label or a duplicated region.

Allowing duplicates is what makes the search work: with duplicates forbidden, greedy growth of the same quadrangulations reaches 98.3% of the labels and jams, because the last lens insertions require two missing labels to coincide geometrically. With duplicates allowed, a missing label can be created next to a duplicate at zero cost and defects diffuse until they annihilate.

17 curves. The walk starts from the Griggs–Killian–Savage diagram [9] for $n = 17$ with every multiple crossing resolved into simple crossings, a state with all labels present and 69,394 duplicate regions. With λ ramped from 0.3 to 1 over three hours at $T = 0.25$, three of fifteen independent runs reached $E = 0$; a fourth diagram was obtained by a further 40-minute cycle from a state with 68 duplicates. Ten later instrumented runs of the same recipe finished seven times within three hours, first reaching $E = 2n$ after $1.6\text{--}2.8 \times 10^9$ proposals and finishing within 1.8×10^9 more. The same method finds 11- and 13-curve diagrams from the trivial starting state in seconds and minutes: on a laptop busy with other work, 200 of 200 seeds produced a simple symmetric 11-Venn diagram, in a median of 0.8 seconds and at most 16 seconds, all 200 pairwise non-isomorphic (up to cyclic relabelling, mirror image and swapping the poles) and all non-monotone; 11 of 13 seeds produced a 13-curve diagram within a ten-minute limit, in a median of 52 seconds, again all distinct and all non-monotone.

19 curves: budget. The resolved starting diagram for $n = 19$ has all labels and 305,862 duplicate regions. Scaled by the number of regions, no 19-curve run in the first three days received the budget that the 17-curve runs needed: the longest had 6×10^9 proposals and was still descending when stopped, and every restart from a low-energy state was shorter than the 17-curve endgame itself.

19 curves: the knot. Every lineage that reached low energy stalled with the same duplicated label, $\{0, 3, 4\}$, whose two instances had degrees 6–9 and shared a neighbour. A region of the dual can move by a triangle flip only at degree 3, so the walk’s targeted proposals, when they landed on such a duplicate, did nothing. The duplicate is inherited: in the resolved starting diagram the label $\{0, 3, 4\}$ already appears twice, once as a region of degree 18. (The resolved 17-curve diagram has the same feature for $\{0, 1, 4\}$, degrees 4 and 17, and the 17-curve walk happens to grind it down; the ten 17-curve pre-completion states all have their duplicates at degree 3 or 4.) One added proposal class removes the obstruction: when a targeted proposal lands on a duplicate of degree above 3, it attacks that degree, by a triangle flip at a degree-3 neighbour or a lens insertion on an incident edge, both ordinary moves under the ordinary acceptance rule. From a frozen $E = 38$ state, 25 minutes at $\lambda = 0.35$ with this proposal reduced $\{0, 3, 4\}$ to multiplicity one; without it, 45 minutes at $\lambda = 0.30$ left it untouched. A three-hour ramp $\lambda : 0.5 \rightarrow 1$ then cooled the state to $E = 38$ with two missing labels of rank 11 and no duplicates.

19 curves: closure. From that state, continuations differing only in the random seed were run with λ held at 1 and with λ ramped from 0.6, 0.8 or 0.9 to 1. Held at 1, five of eight continuations reached $E = 0$, after 4 seconds, 13 seconds, 190 seconds, 4.3 hours and 6.5 hours; of the sixteen ramped continuations one finished, at the end of its six-hour ramp. The pre-completion states are saved exactly, by journal undo of the winning move inside the engine, and show three closing mechanisms: a single triangle flip of the last duplicated region, of degree 3, into the last missing label at Hamming distance exactly 3 (four diagrams); a single lens insertion creating both missing labels at once after they had drifted to Hamming distance 1 (one diagram); and a single lens removal of two adjacent duplicated regions in a state with no missing label (one diagram).

5 Properties

All ten diagrams are non-monotone in the sense of Bultena, Grünbaum and Ruskey [6]: in the first 19-curve certificate, 3,590 of the 27,594 region orbits (13%) lack a neighbouring region of rank one lower or one higher, and the 17-curve certificates behave the same way. Since monotone diagrams are exactly those that can be drawn with convex curves [6], none of the diagrams reported here has a convex drawing. Every previously published simple symmetric Venn diagram with $n \geq 11$ is monotone, and the crossing-sequence representation used to find them [15, 14] enumerates monotone diagrams by construction, so the diagrams reported here were outside the search space of those methods; Mamakani and Ruskey themselves noted that the size of their search sequences grows about tenfold from 13 to 17 curves [15]. Minimum region degree is 3 (a degree-2 region is impossible in a simple Venn diagram with $n \geq 3$); in the first 19-curve certificate the degree histogram is 187,340 regions of degree 3, 199,120 of degree 4,

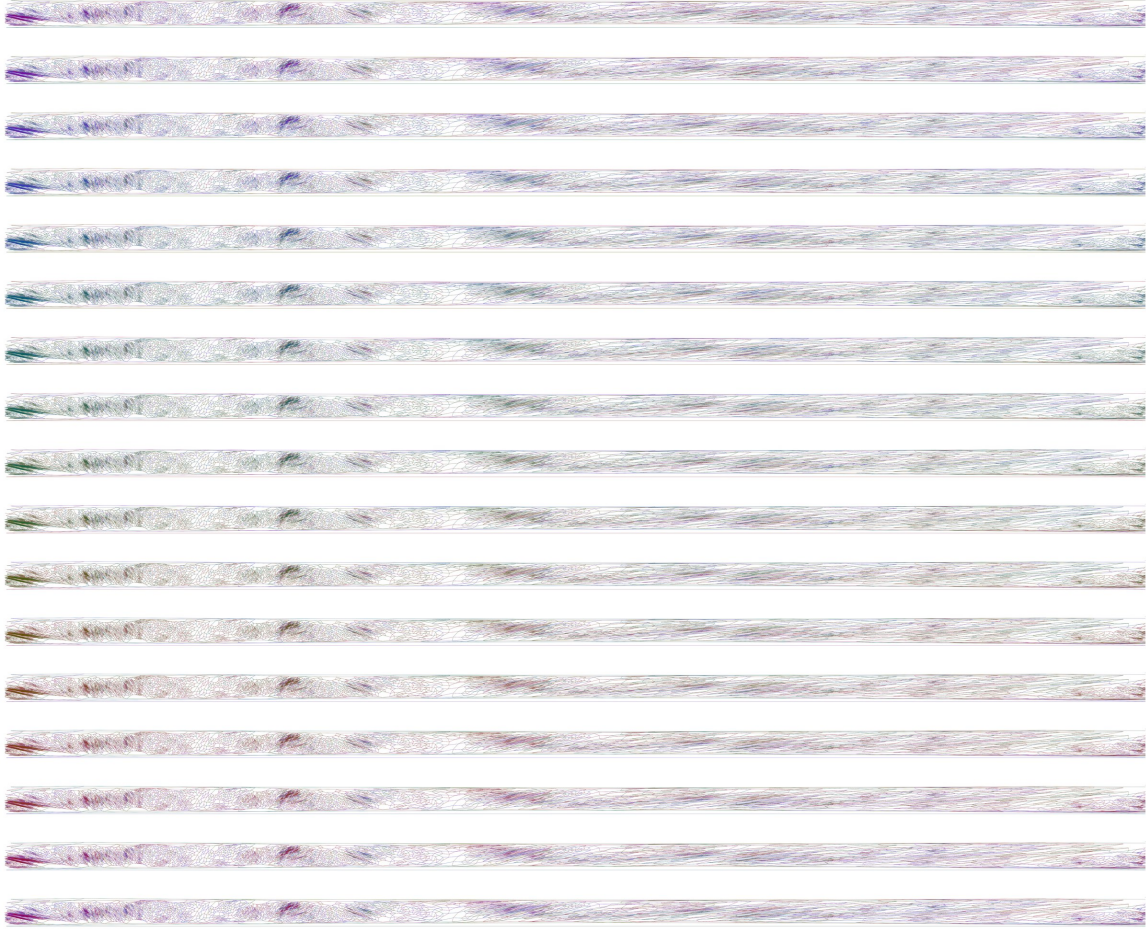


Figure 2: The 17-curve diagram `c178d7bd` unrolled: its seventeen congruent sectors, each a fundamental domain of the rotation containing $131,070/17 = 7,710$ crossings, drawn as rows, with the colour of each curve advancing by one position from row to row. The whole diagram is one row bent around the centre seventeen times.

98,686 of degree 5, and a tail to degree 16, with the two poles of degree 19 and every rank-1 and rank-18 region of degree 11.

6 Data

Certificates, pre-completion states, logs, the search code with its checksums, the independent checkers and both Lean developments are at <https://github.com/dzoba/venn17> and archived on Zenodo (DOI to be inserted).

Tool and computational resource disclosure

This work was carried out by two AI systems working under the author’s direction over five days, and this note discloses their roles in full, in the spirit of the Leiden Declaration on Artificial Intelligence and Mathematics. Claude (Anthropic) formulated the dual-map encoding, designed and implemented the search (the relaxed walk, its moves, energy and annealing schedules), diagnosed the failure at 19 from the run records, designed and implemented the added proposal class and the capture of pre-completion states, ran the searches and the verification, performed the analyses reported here and drafted this note. Codex (OpenAI) constructed the resolved Griggs–Killian–Savage starting states, wrote independent validators for every certificate, identified the need for the bigon moves, built exact local solvers used to diagnose the earlier plateau, and ported the Lean development to 19 curves. The author set the goals and strategy, decided what to pursue and what to abandon, supplied the computing resources, and is responsible for the content. Computing: one 16-core laptop throughout, and for the 19-curve runs cloud instances of up to 24 vCPUs each, never more than about a hundred vCPUs at once and none for longer than two days; total cloud cost under five hundred dollars. Software: the search engine, checkers and plotter in the data repository (C++ and Python with NumPy); Lean 4 with Mathlib for the formal proofs; `tectonic` for this document.

Acknowledgments

We thank Justin Grimes for the formal verification of the 17-curve certificate in Lean, carried out independently and without knowledge of our construction, on which the 19-curve verification is built. We thank Khalegh Mamakani for his encouragement from the first day, for pointing out that the earlier searches excluded non-monotone diagrams by construction, and for his advice on how this work should be presented; Stan Wagon for correcting the history given here, for the references, and for his and Peter Webb’s repair of Henderson’s proof; and Carla Savage for her kind reply and, with Jerrold Griggs and Charles Killian, for the construction from which every one of these searches begins. The survey of Ruskey and Weston was our map of the field throughout.

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